Multi-VLAN Aruba Switch Project (Hands-On)

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Project Overview

This document provides detailed guidelines for configuring DHCP on an Aruba 6300 switch to establish a local network for three clients. The setup includes creating three VLANs, each assigned to a client, with DHCP dynamically distributing IP addresses from a specified range. Additionally, all clients must have internet access and be able to communicate with one another through the switch. Communication between the clients will be secured using SSH with key-based authentication.

Key word definitions for this project

VLAN: Virtual Local Area Network(s) which are used to segment the physical network. We use this to create different networks for each end user (Our PC).

SSH: Secure shell. We use this to access our accounts from one VM to another using our generated key.

IP Routes: Defines a path that network traffic can travel to. We use this to allow our VM's to communicate to each other since we are in three separate VLANs.

DHCP: Dynamic Host Configuration Protocol. This assigns an IP automatically to whomever is at the end. We use this to get an automatic IP that we can use for our VM's.

Subnet Determination for Given Student IDs

In this project, the subnet is defined as **172.16.x.?/26**, where **x** represents the unique ID assigned by random. However, determining the correct subnet involves more than simply replacing **x** with the ID. Instead, we must calculate the appropriate values using the **/26** subnet mask.

Each IP address consists of 32 bits, with **172.16** being fixed and occupying the first **16 bits**. The **/26** subnet mask leaves us with **10 bits** to define our subnet.

To determine the subnet for a given student ID:
1. Convert the **student ID** to a **10-bit binary representation**.
2. Split it into **8-bit + 2-bit** sections.
3. The first **8 bits** form the third octet of the IP address.
4. The remaining **2 bits** contribute to the starting address in the fourth octet.
5. By setting the last **6 bits** to **0**, we determine the first address in the subnet.

Subnet Calculations

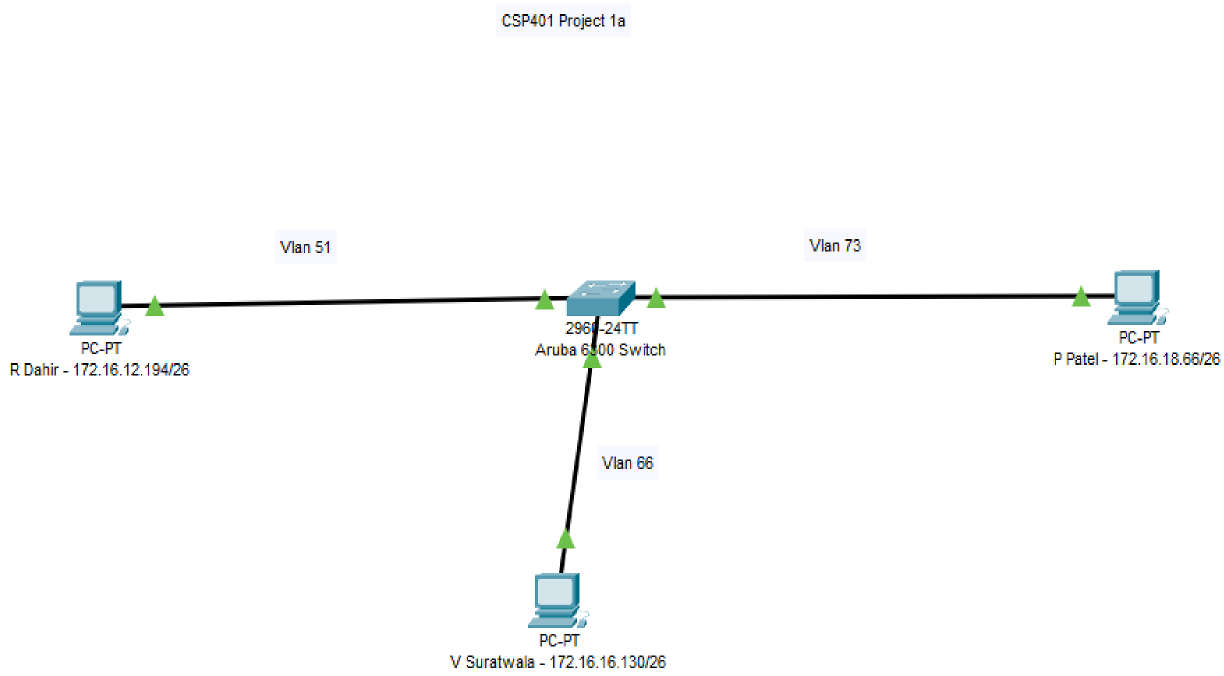
1. **Client1 (ID 73) PR**
• **Binary**: 0001001001
• **Splitting**: 00010010.01??????
• **Converted to Decimal**: 18.64
• **Subnet**: **172.16.18.64/26**

2. **Client2 (ID 66)**
• **Binary**: 0001000010
• **Splitting**: 00010000.10??????
• **Converted to Decimal**: 16.128
• **Subnet**: **172.16.16.128/26**

3. **Client3 (ID 51)**
• **Binary**: 0000110011
• **Splitting**: 00001100.11??????
• **Converted to Decimal**: 12.192
• **Subnet**: **172.16.12.192/26**

Thus, the assigned subnets are:
• **Client1**: **172.16.18.64/26**
• **Client2**: **172.16.16.128/26**
• **Client3**: **172.16.12.192/26**

Network Topology



Implementation

Step 1: Setup the physical topology

The first thing done when trying to connect to the switch is making sure everything is plugged in to the correct ports, and that we can connect to the switch. Some things done were:

- Knowing which port was being used for on our computer (G1, G4 etc.). Need to make sure that our port number aligns with the port of the switch. If we didn't have the correct port number, we could not have access to the switch.

- Setting up cables in the back. To make sure that our switch can communicate to our personal computers and VM's, we need to connect the cables into the correct ports.

Step 2: Access To The Switch

The second thing is to make sure we can SSH from the computer to the switch. This can be done by using the command: ssh <name>@<switch-ip-address>. After met with the password prompt, no password is needed, and all we need to do is press enter. Now in User Executive Mode.

Step 3: VLAN Configuration

The network topology we were provided with showed that we needed to have three separate VLANs for our three VMs. This is how it was done:

- In configuration mode, using the command "vlan <number>" creates a VLAN for you. For us we created 3 different VLANs.

- Since the users within this VLAN are going to be communicating with users outside the VLAN, it needs to have a default gateway.

- Each VLAN was assigned to a specific interface, for example interface 1/1/1 was assigned to one VLAN, interface 1/1/2 was assigned to another VLAN etc.

Step 4: Set-up DHCP

Since static IP addresses are not being used in this project, we need to set up Dynamic Host Configuration Protocol (DHCP). This can allow for automatic IP address assignment instead of manually assigning an IP address. We also need to define the pools of DHCP since when the DHCP server automatically assigns an IP, it picks from a pool of available IP's. To do this:

- Use the command "dhcp-server vrf default" to enable DHCP on the switch. It also uses the default VRF instead of a custom one.

- Define the pool. By doing this, you're assigning a range of IP addresses that are available to users of the network.

Step 5: Confirmation On Clients

This step is needed in order to ensure that we are using DHCP, and our VMs can access the internet:

- Each VM should have one adapter set in "Bridged" connecting to the switch. The second adapter should be connected straight to the internet.

- Ensure that the Network adapters are configured correctly. An IP should be obtained automatically, and the ip a command should show us the correct IP address.

Step 6: IP Routes And SSH

IP Routes are needed in order to define a path to a specific network. In this case, we need to route traffic to each VM's network in order for them to communicate. This can allow us to SSH.

- Setup IP Routes for each VM, this can be done in the CLI using "sudo ip route add" command. There should be two ip route commands for each VM.

- Test ping to ensure that the IP Routes were inserted correctly.

Secure Shell (SSH) is now needed in order to access the other VM's. Users are also needed so that each VM can access a designated account.

- Install OpenSSH server using the command: sudo apt install openssh-server

- Each VM should create two users, one for each VM that is going to use SSH. This can be done using the useradd command.

- Each VM should also create a keypair on the new user which is essential for connecting to the client VM's, using the command ssh-keygen. Then copy the key to the user with the command: ssh-copy-id -I [location of public key] [username]@[ip address]

Step 7: Testing Network Configuration

Now that everything is done, we can test all our configuration and ensure that everything works properly.

- Ensure that each VM can be pinged successfully (VM1 to VM2, VM2 to VM3 etc.).

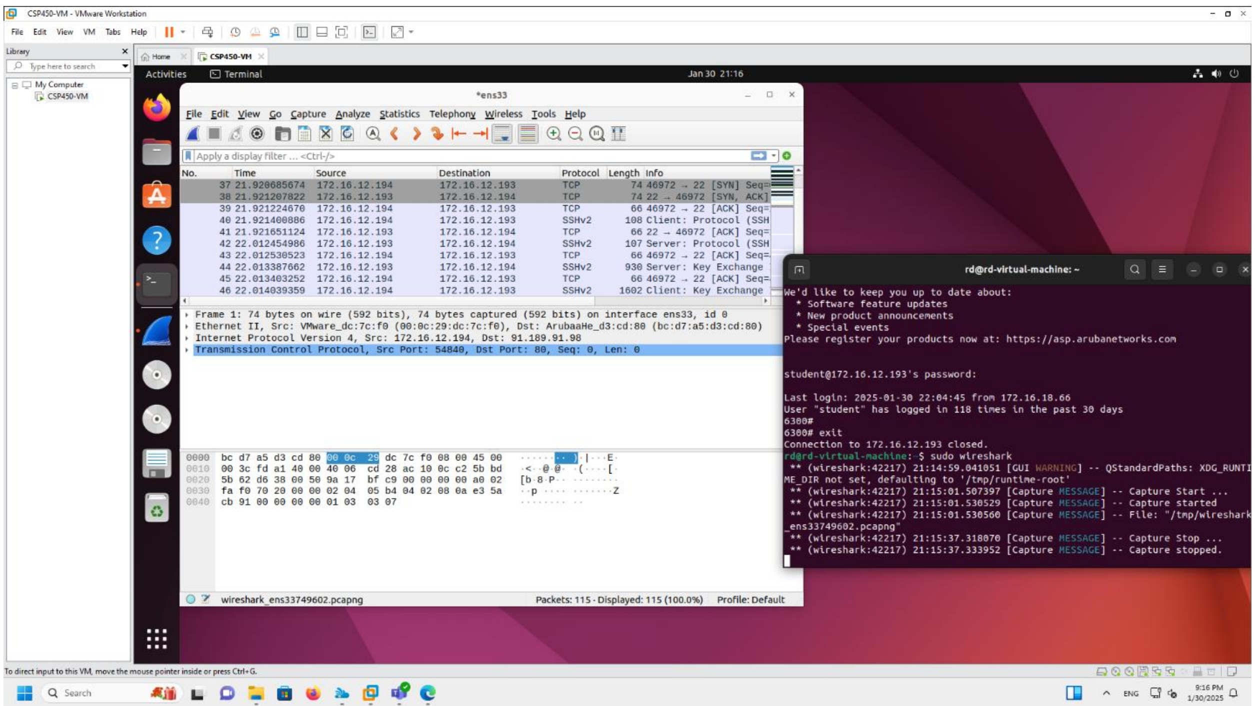
- If you can ping each VM successfully, attempt SSH into each VM. A successful attempt would mean that you can SSH without needing a password.

- Also attempt to SSH into the root account of the user. If you are denied, then that means it was successful.

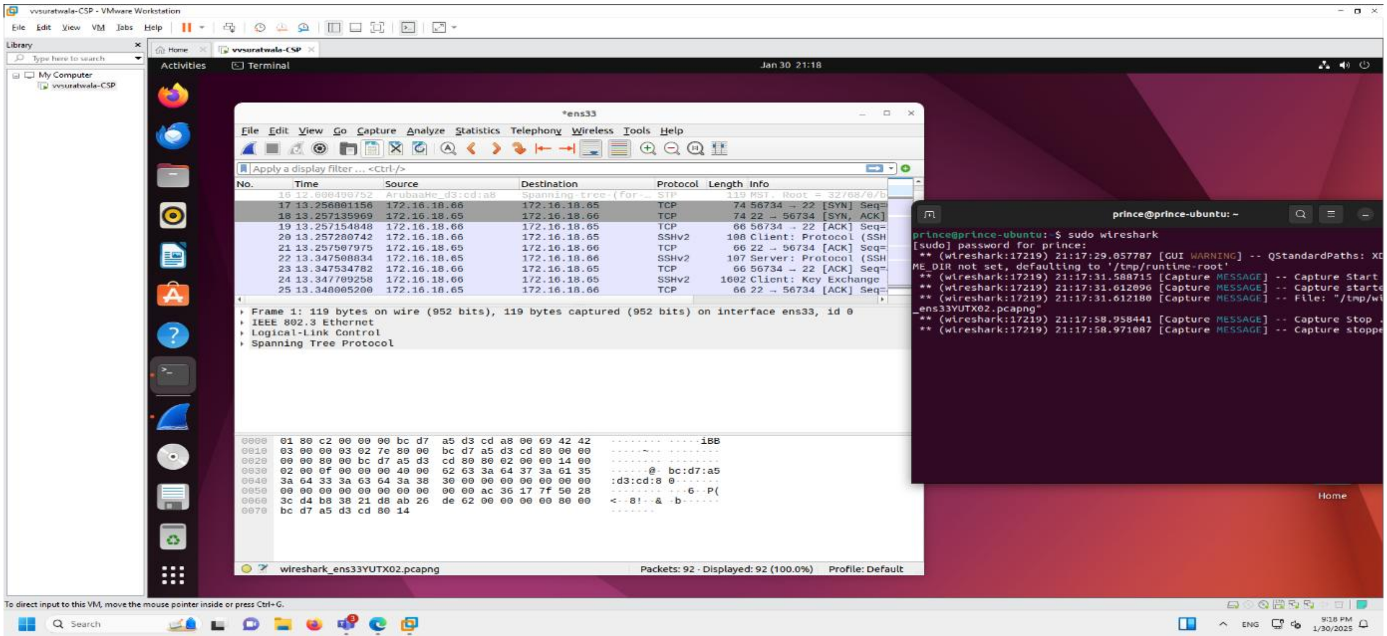
Appendix A: Wireshark

Rooble Dahir: 172.16.12.194 | Prince Patel: 172.16.18.66 | Vrisha Suratwala: 172.16.16.130

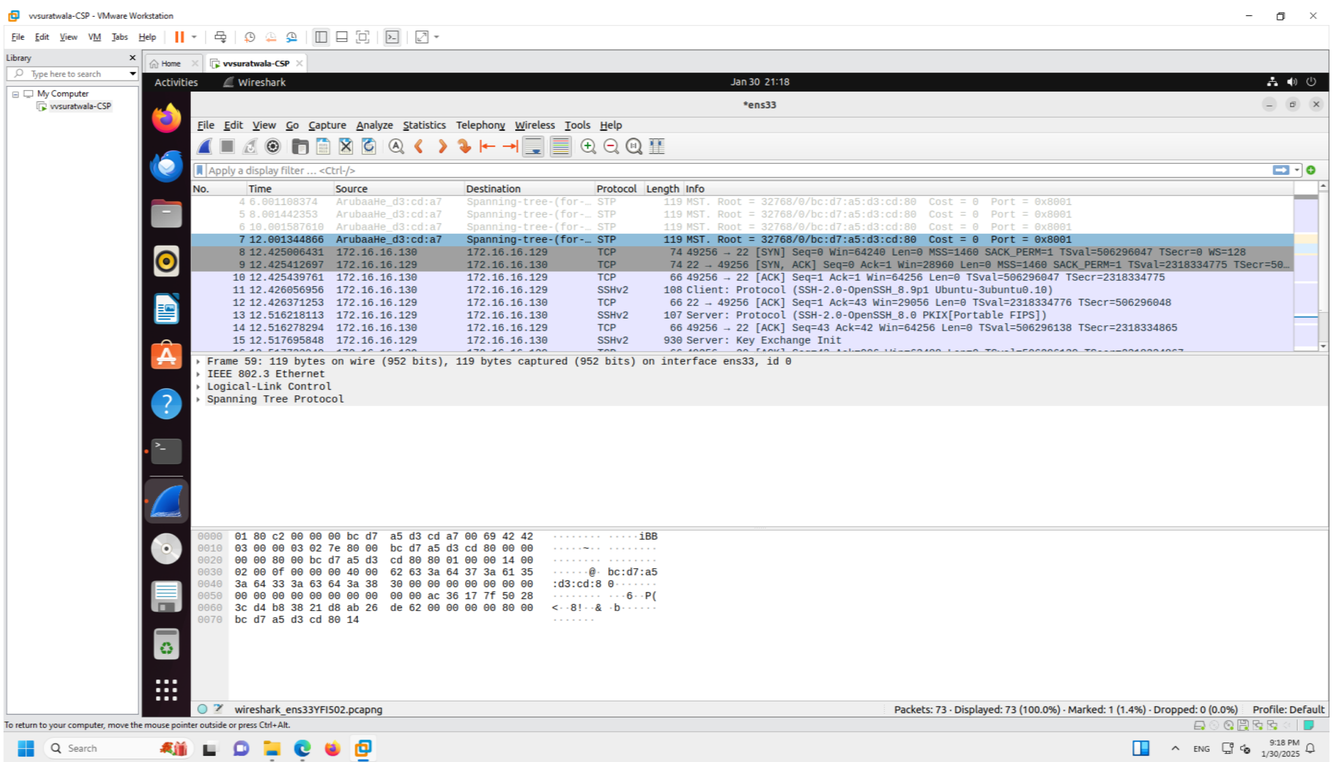
Client3 VM to Switch



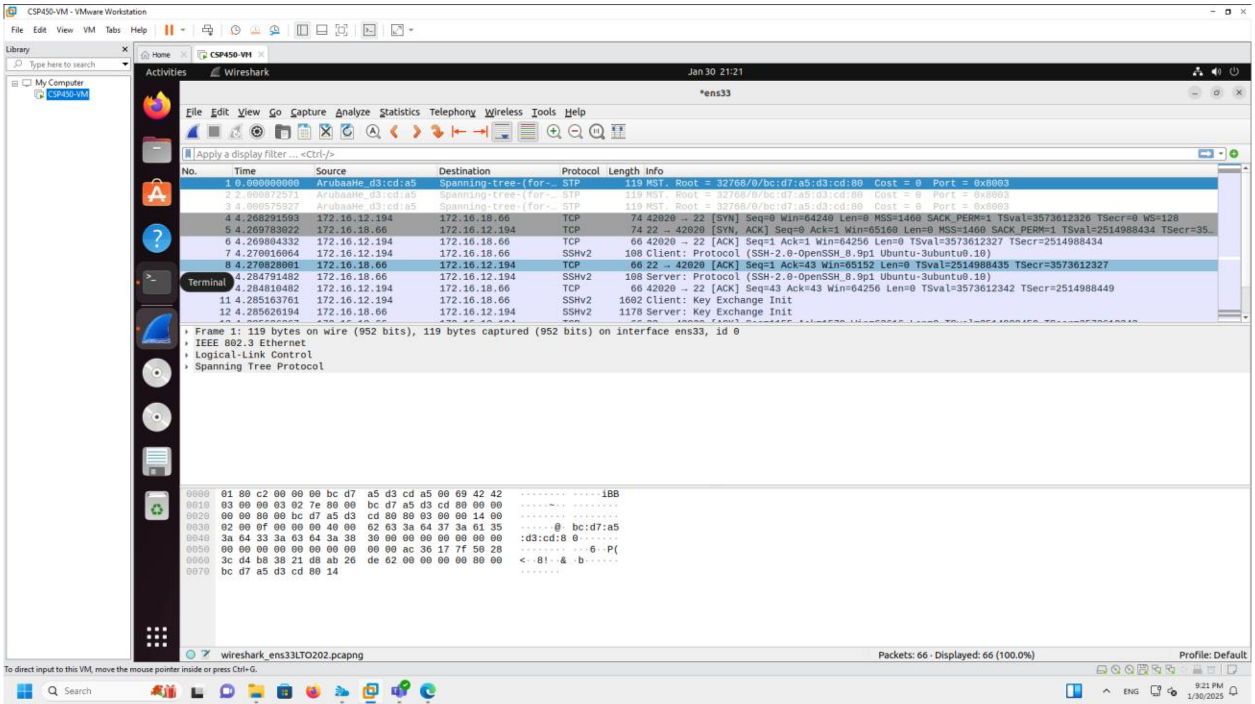
Client1 VM to Switch



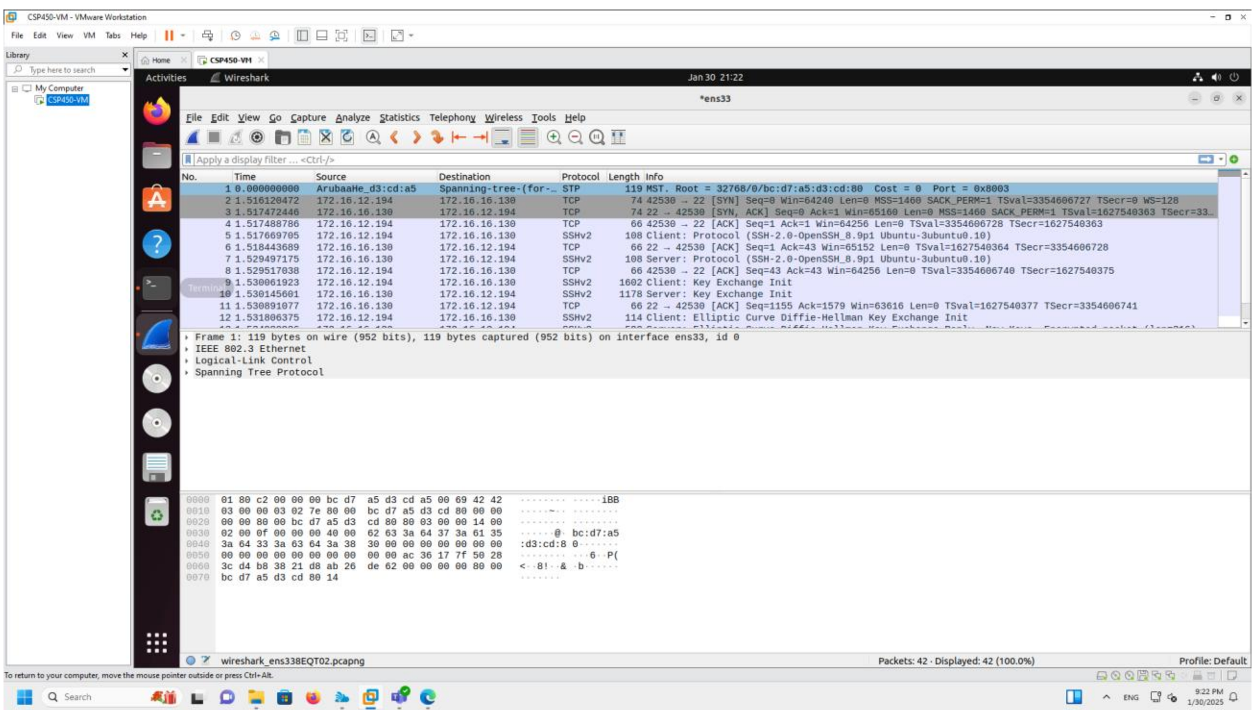
Client2 VM to Switch



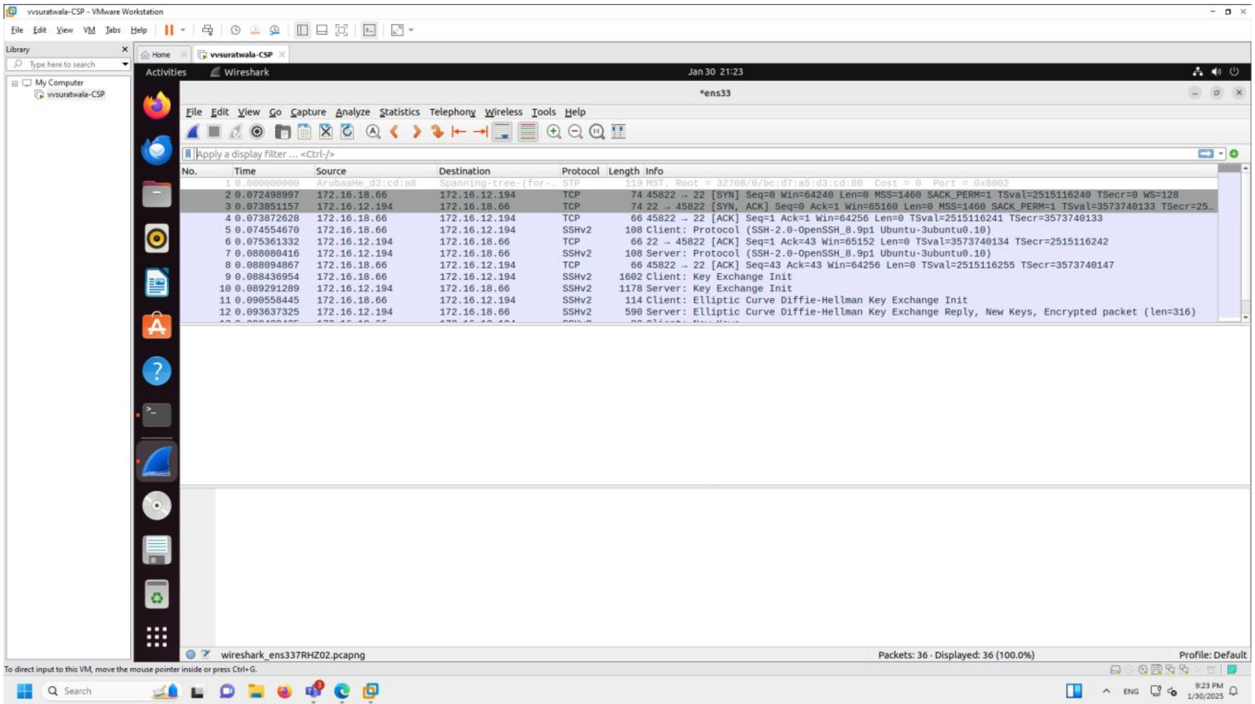
Client3 VM to Client1 VM



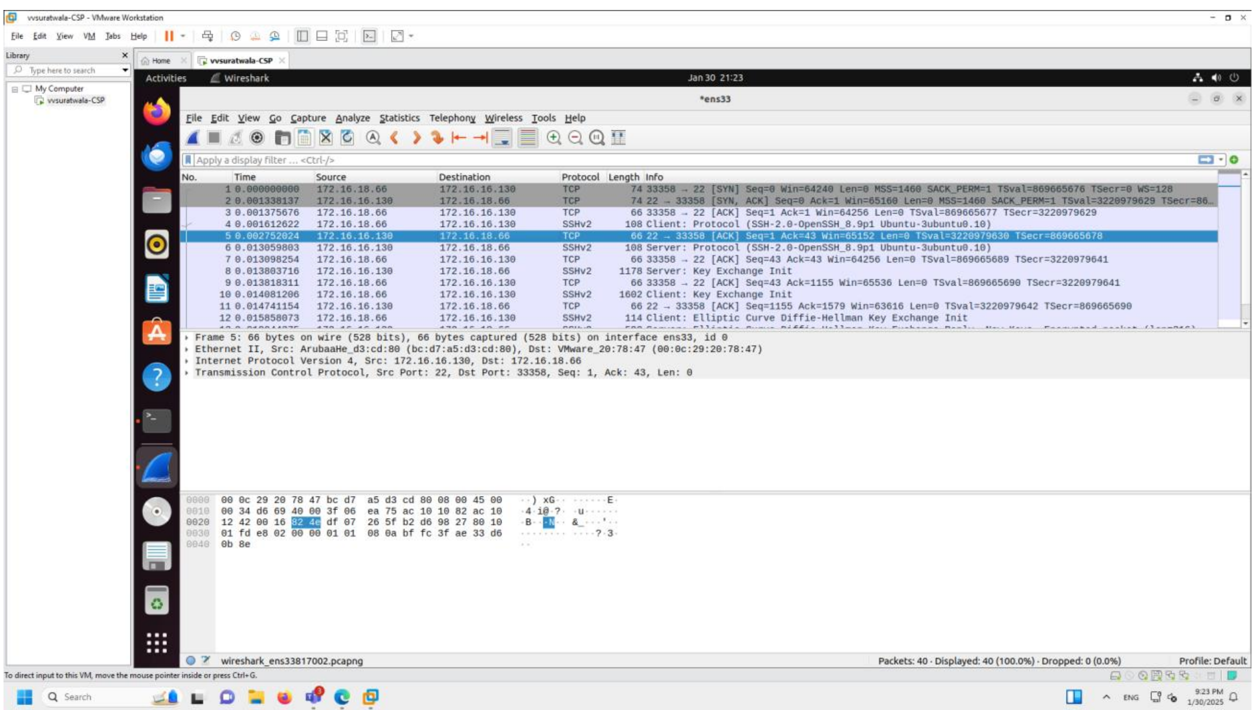
Client3 VM to Client2 VM



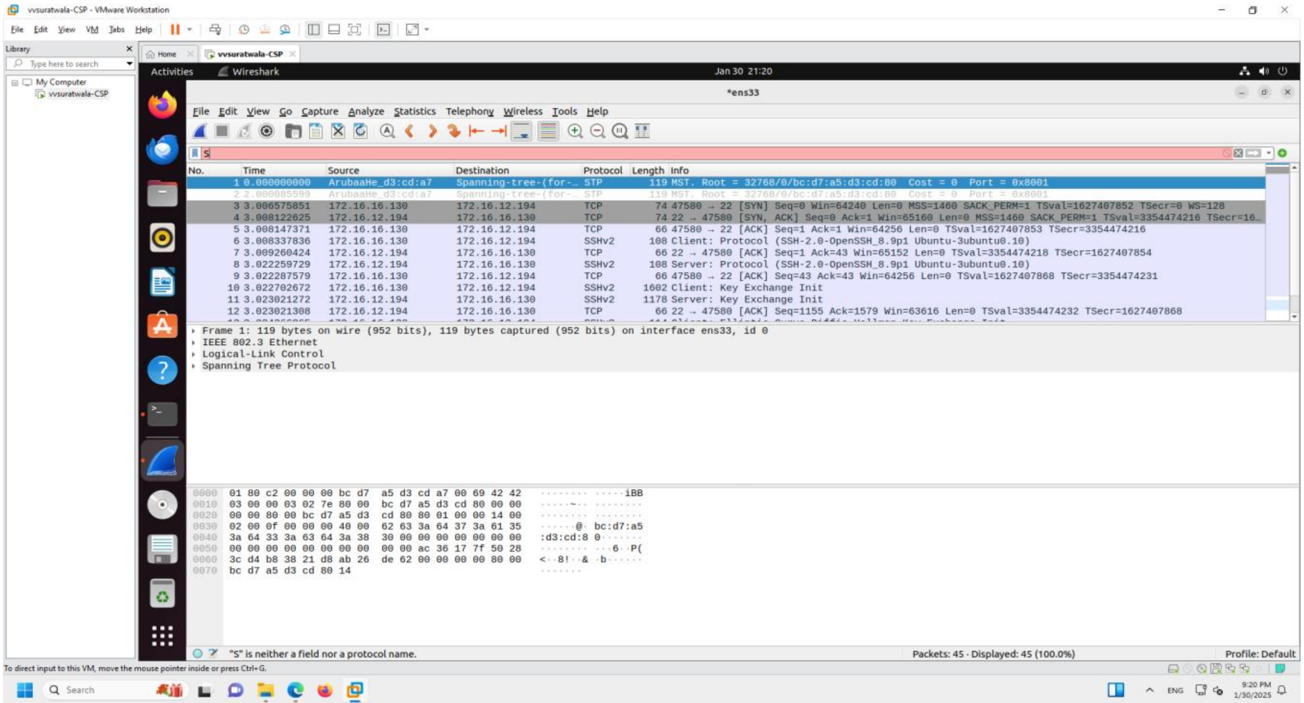
Client1 VM to Client3 VM



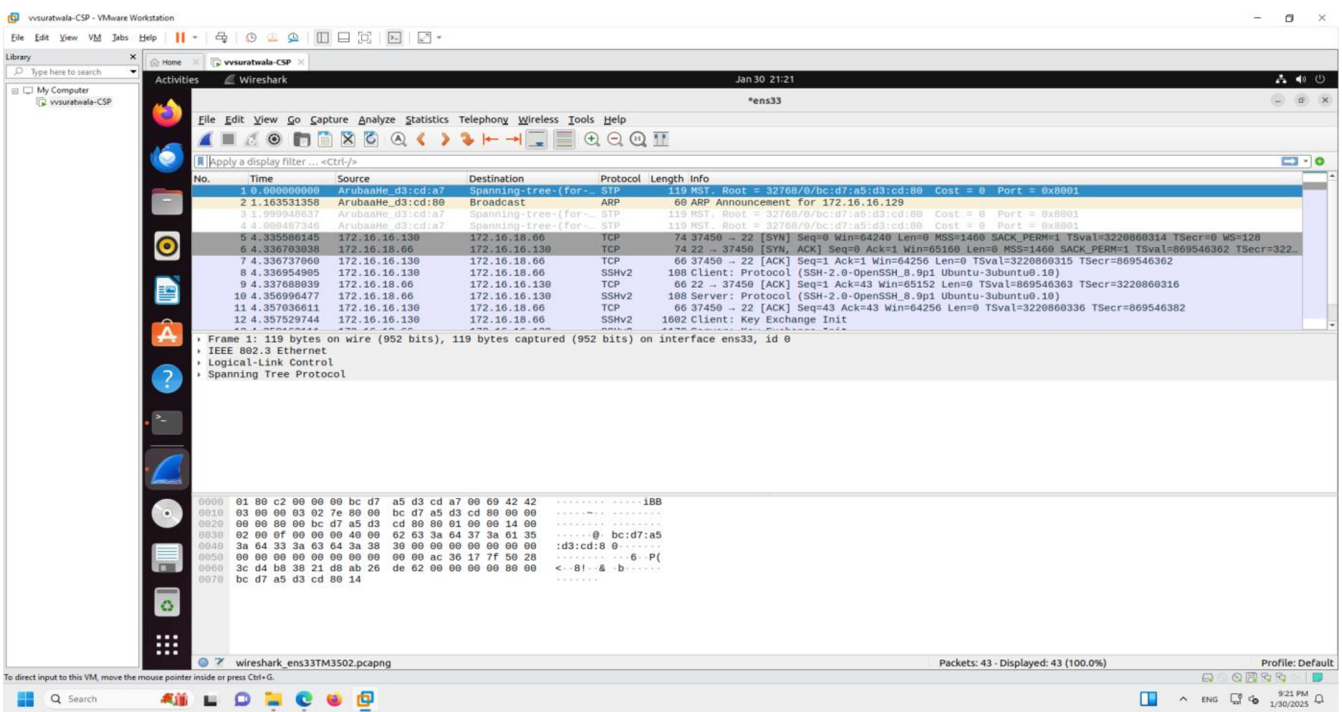
Client1 VM to Client2 VM



Client2 VM to Client1 VM



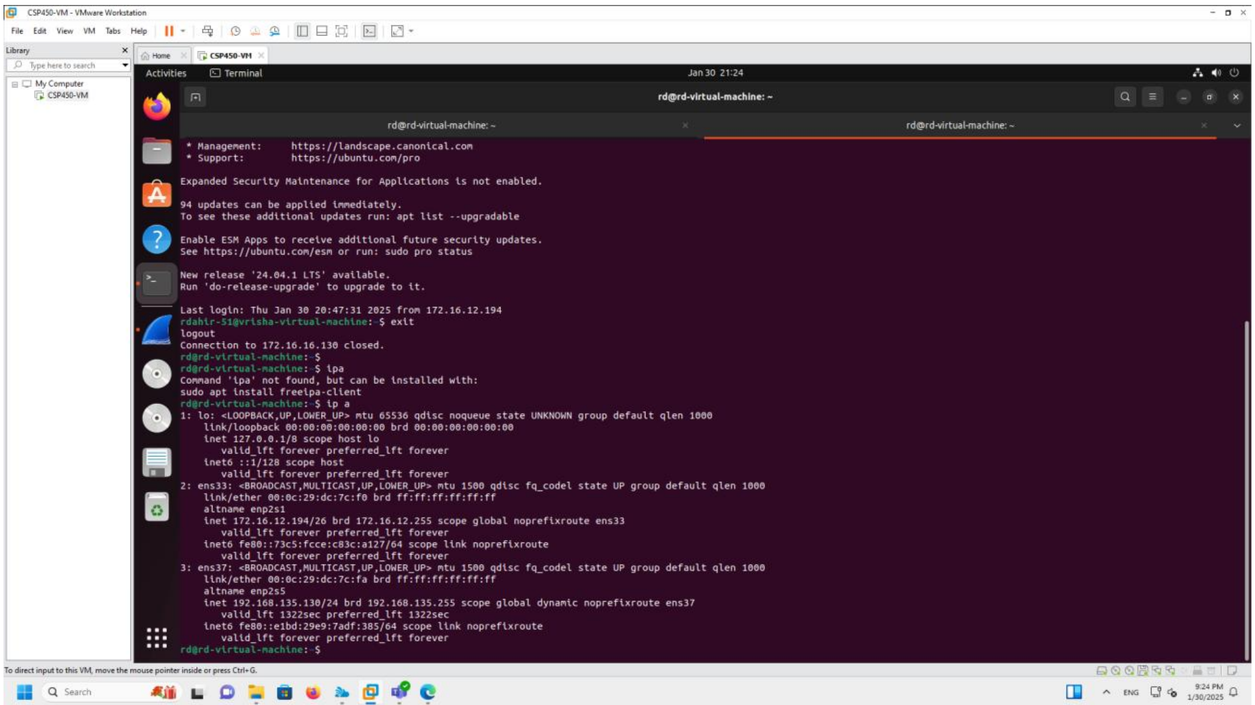
Client2 VM to Client3 VM



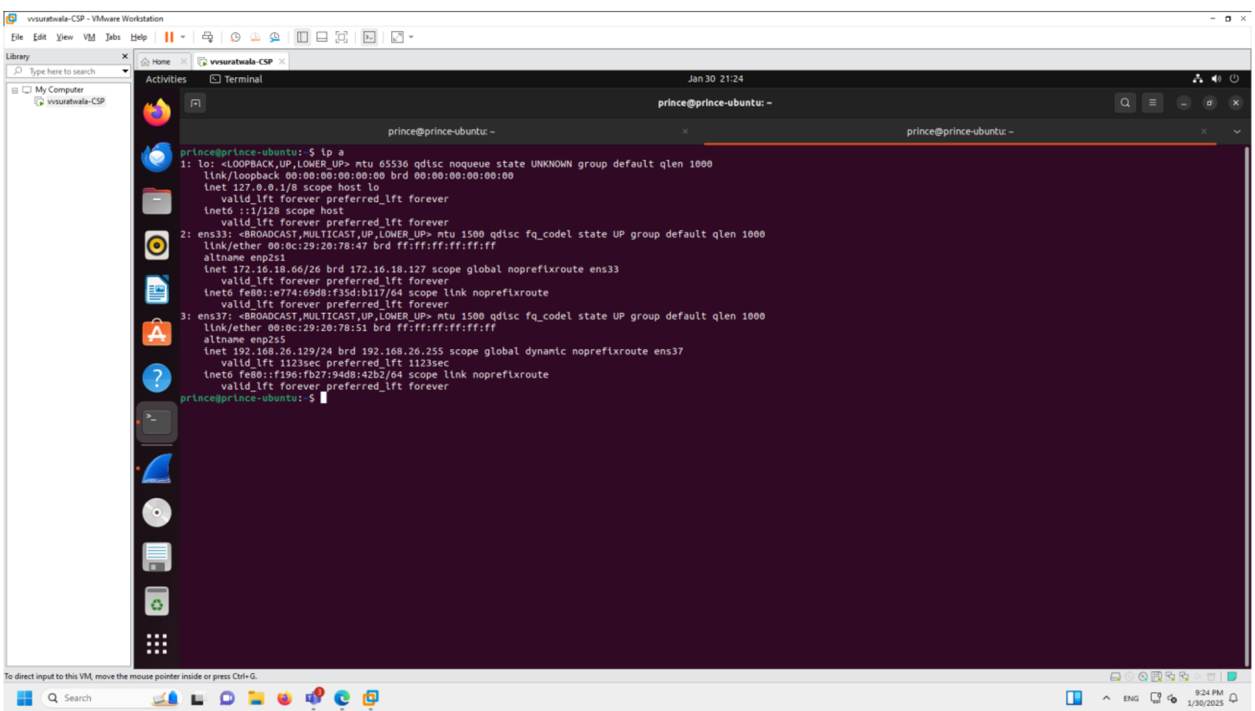
Appendix B: Commands on VM

Terminal Command: ip a

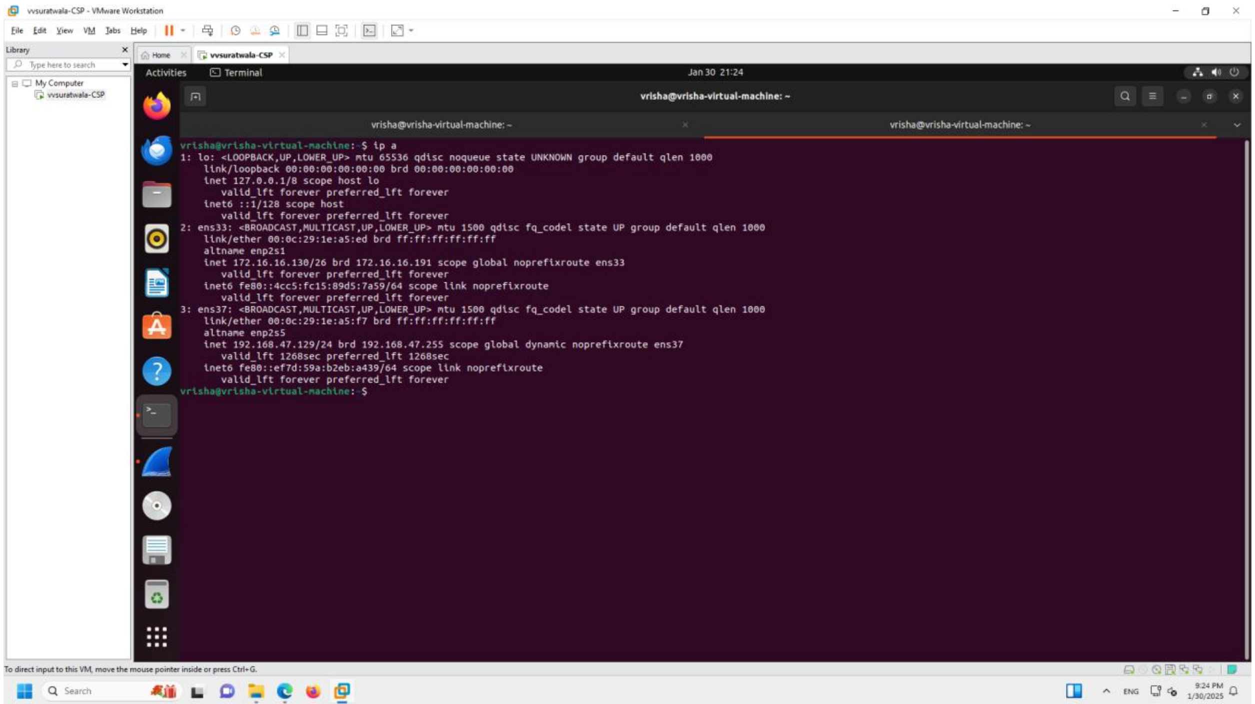
Client1 VM



Client3 VM

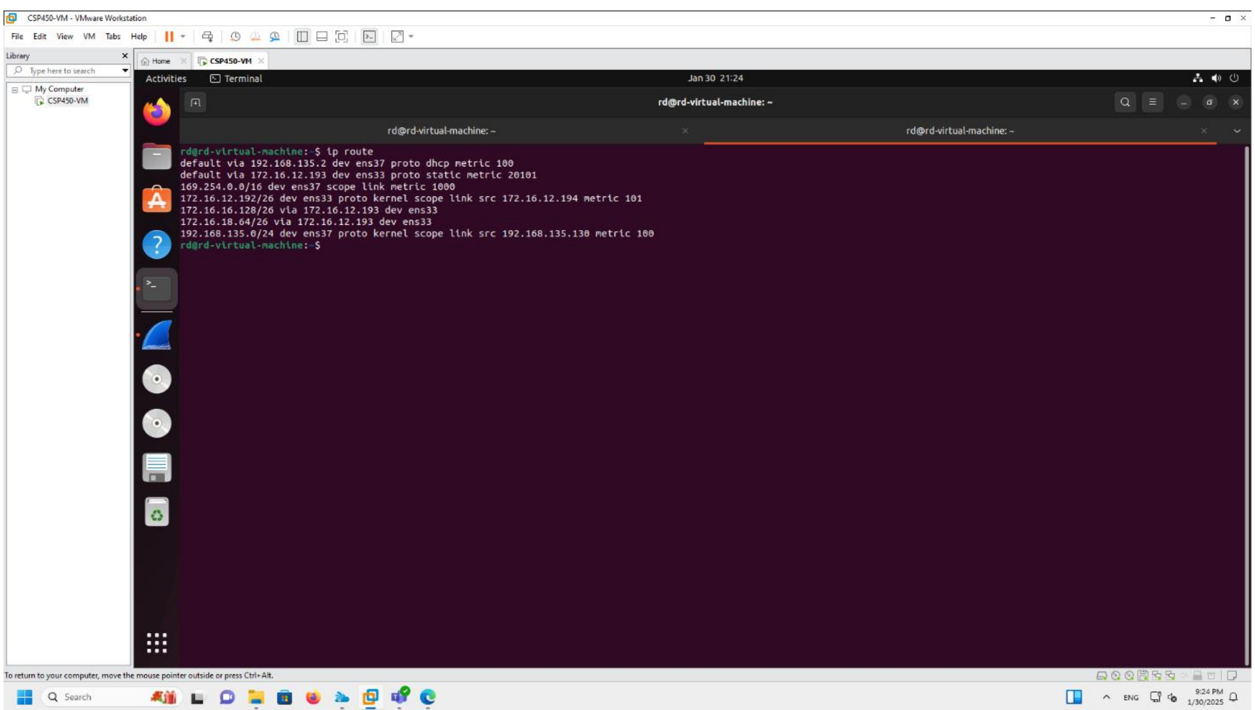


Client2 VM

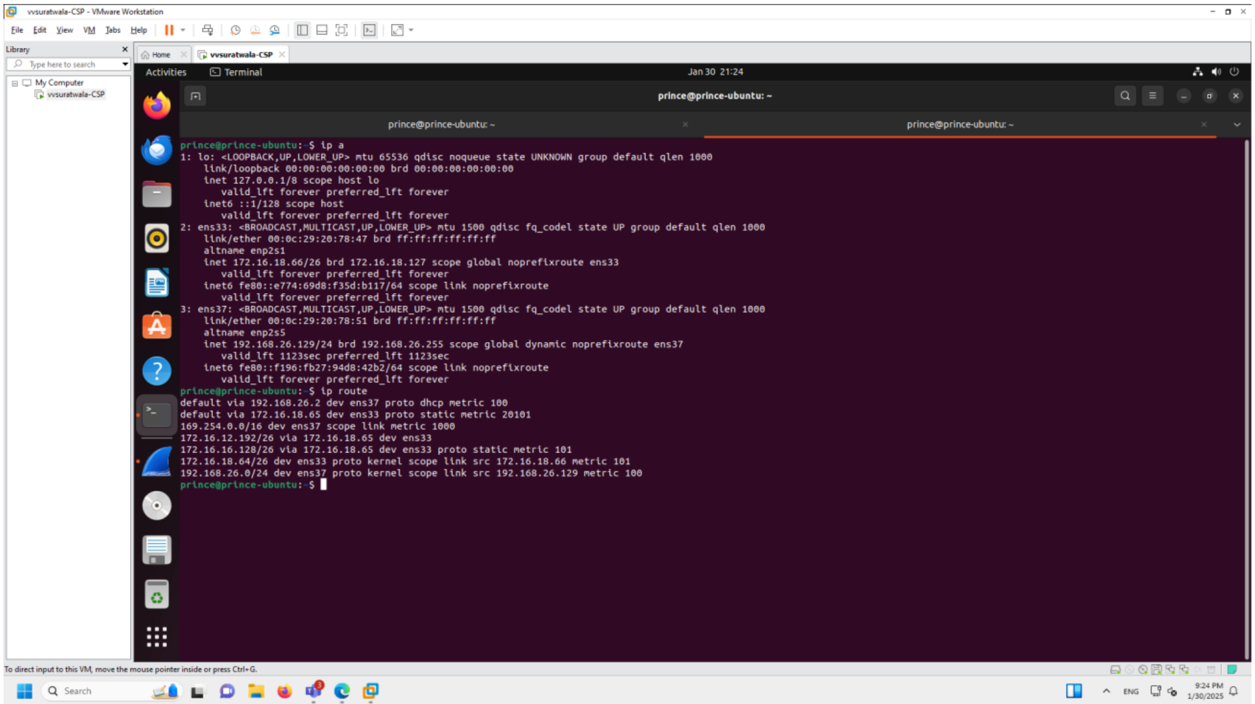


Terminal Command: IP Route

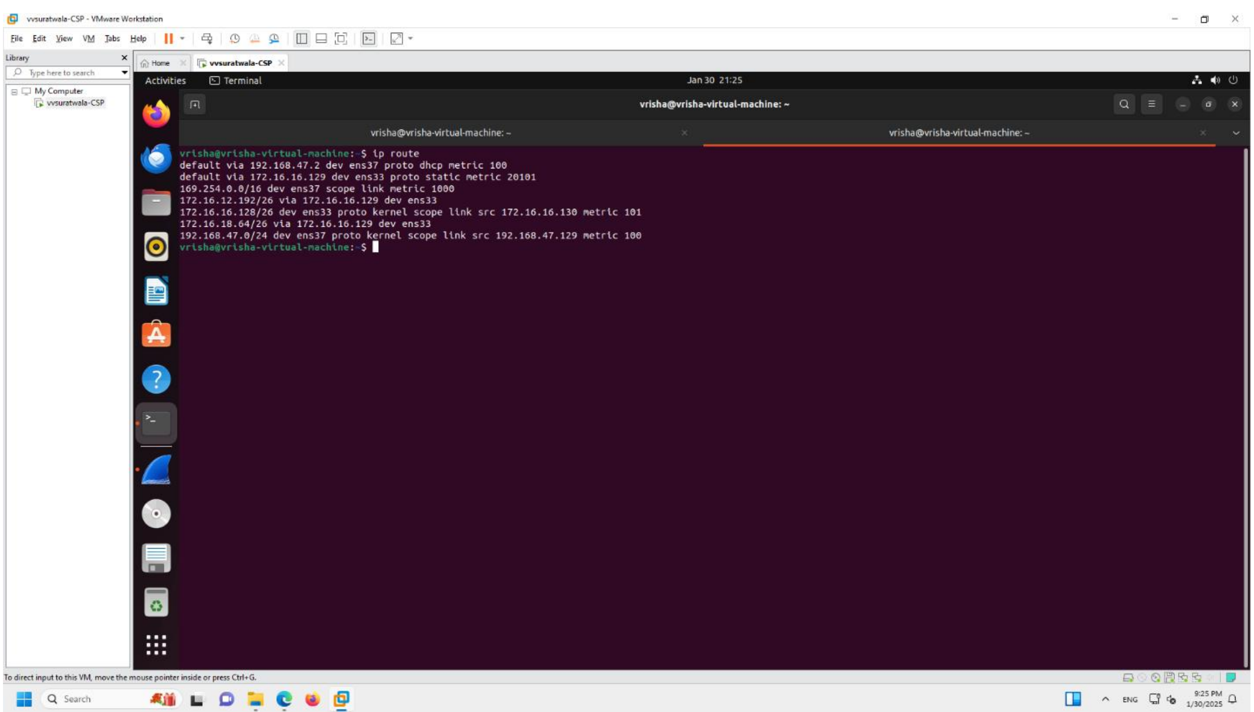
Client1 VM



Client3 VM

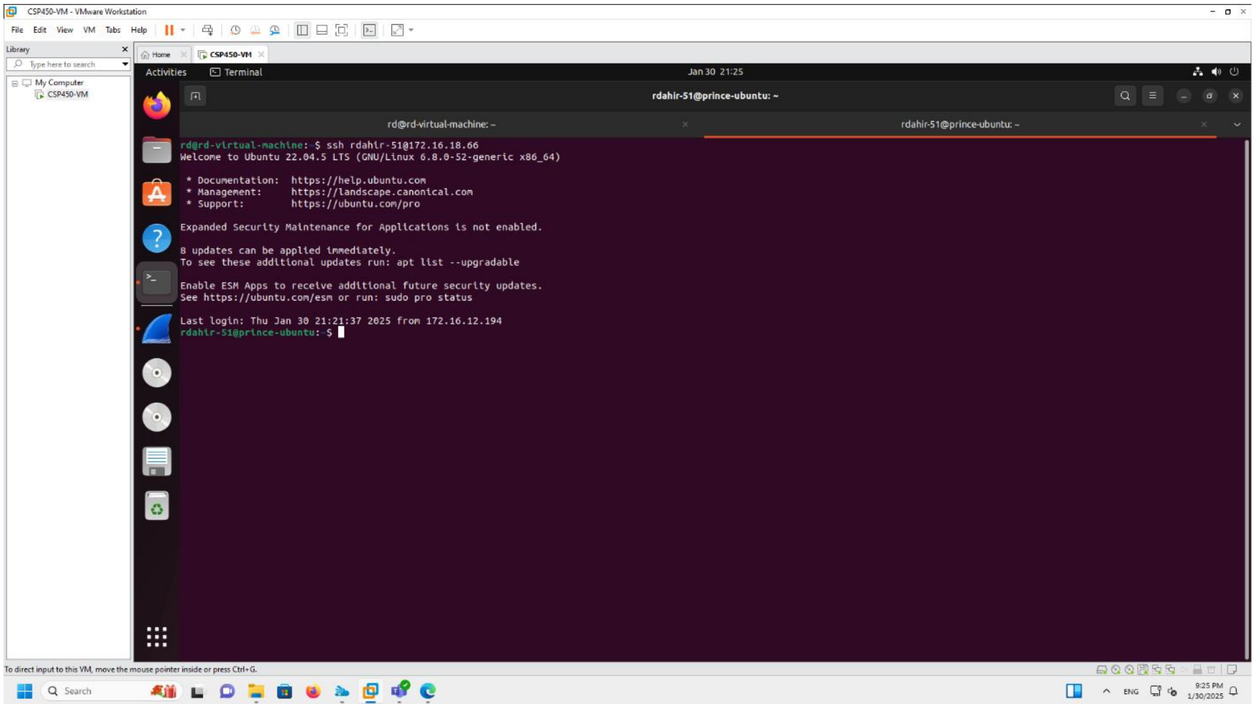


Client2 VM

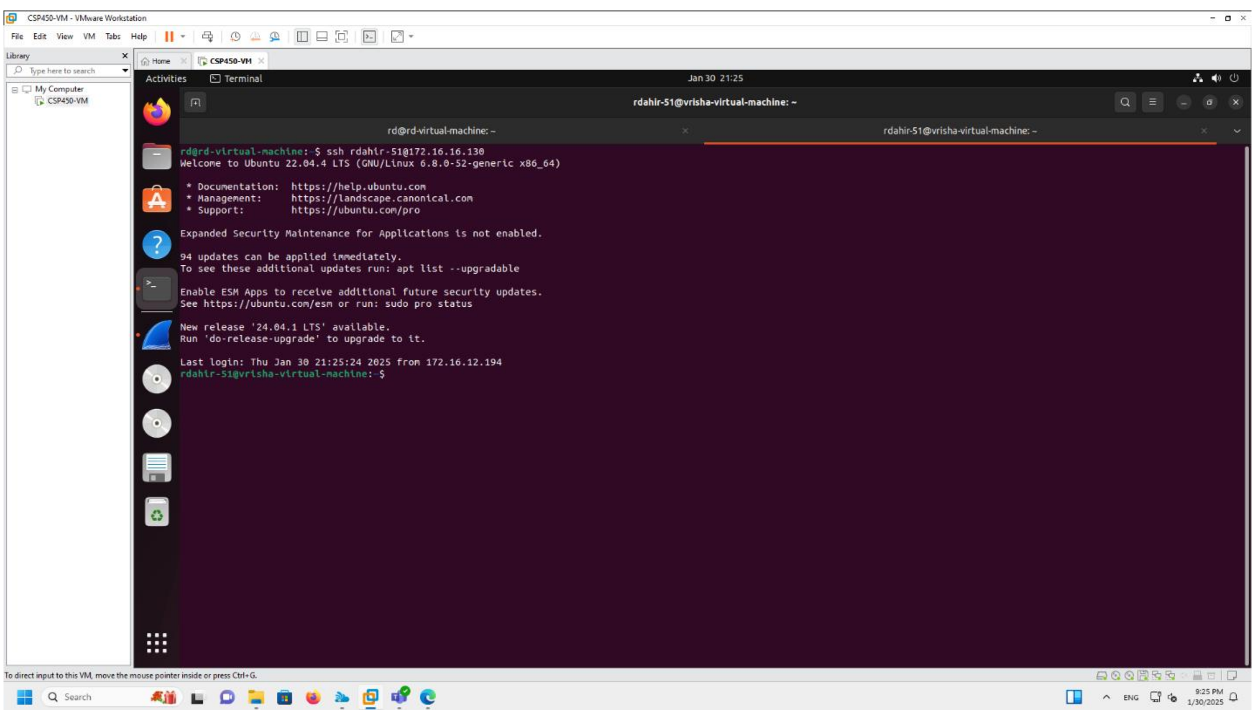


SSH To Partner's

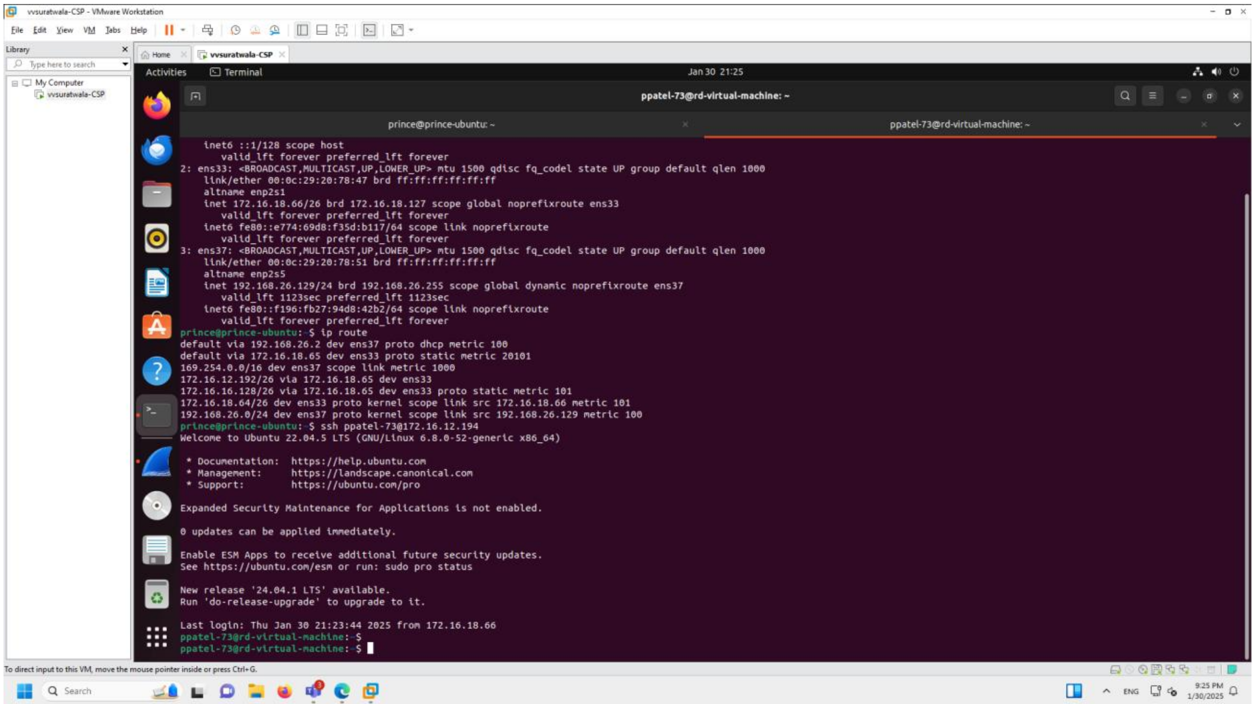
Client1 VM to Client3 VM



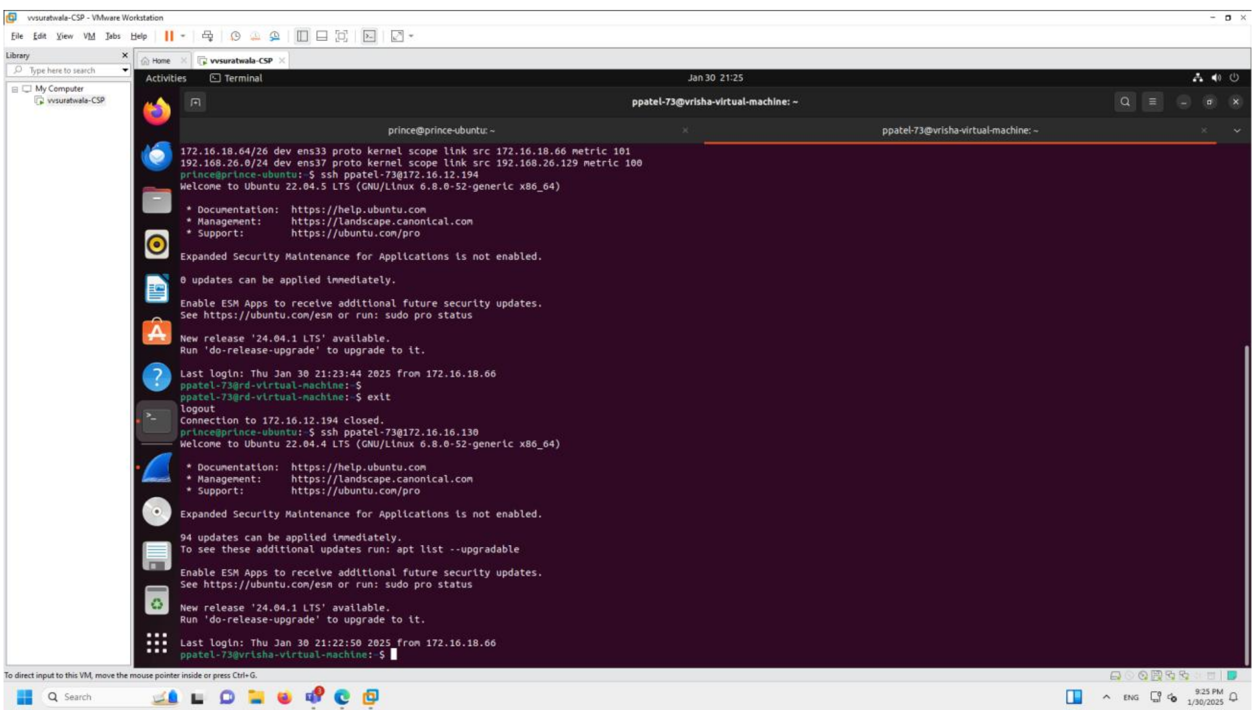
Client1 VM to Client2 VM



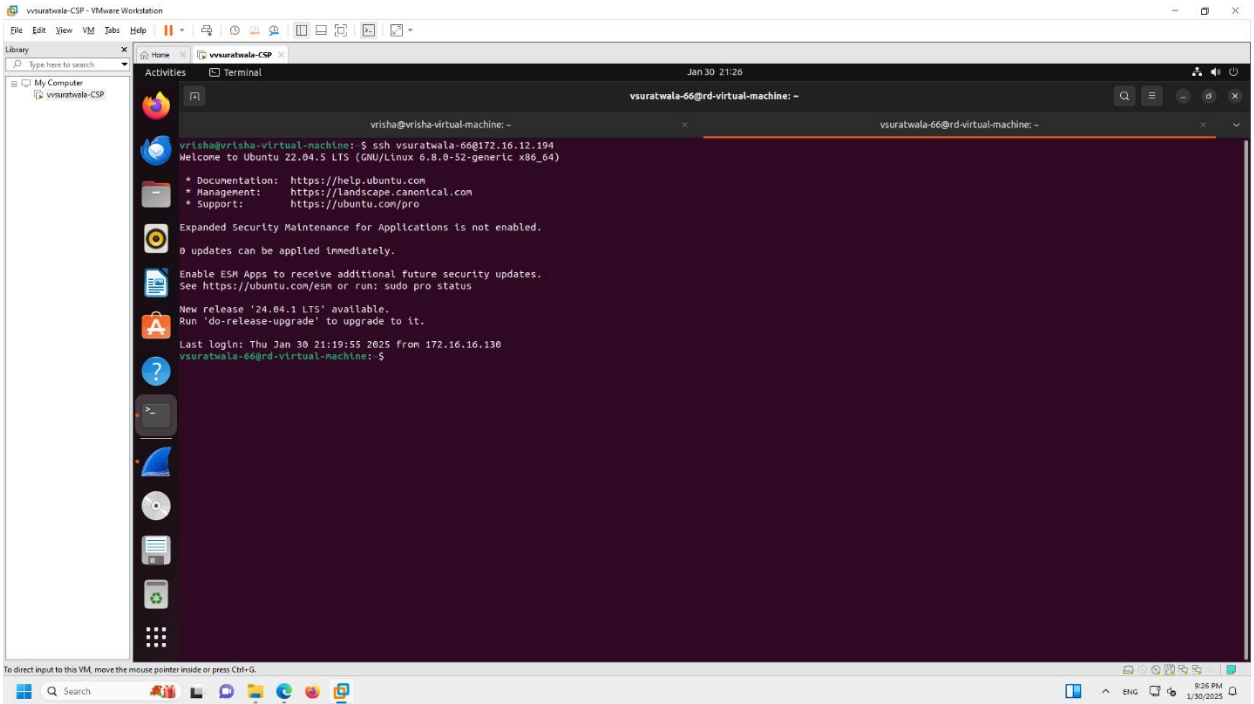
Client3 VM to Client1 VM



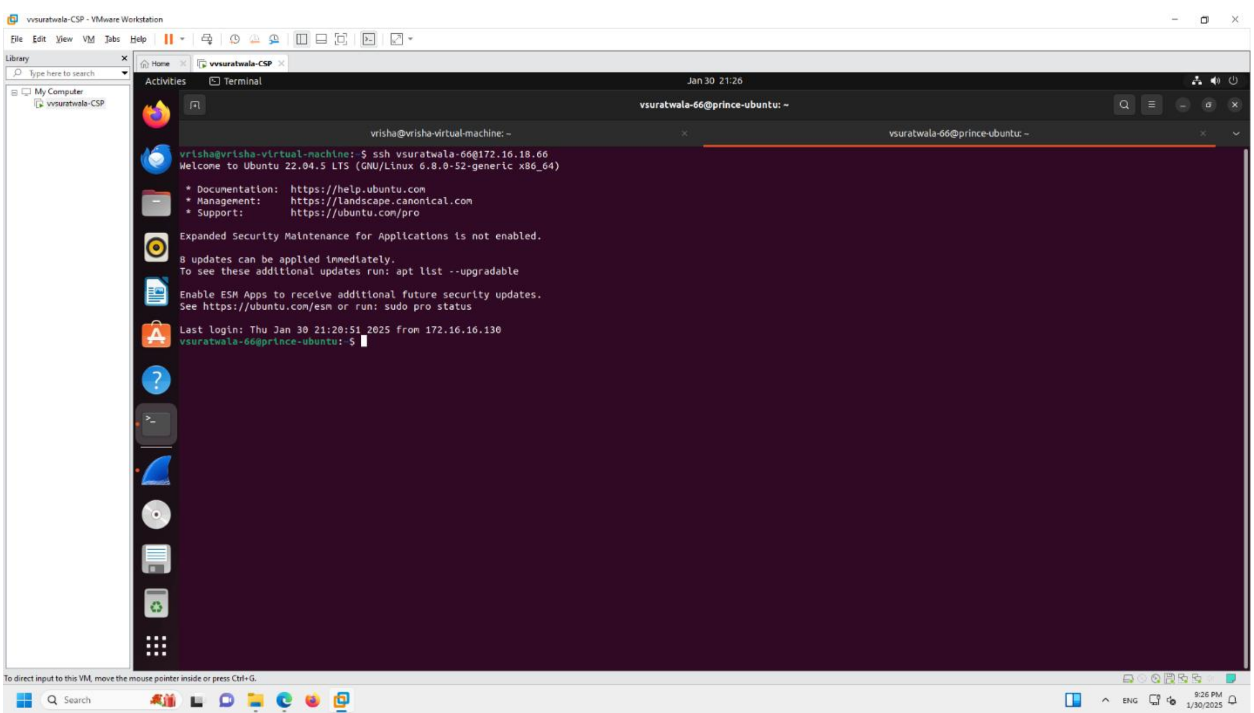
Client3 VM to Client2 VM



Client2 VM to Client1 VM

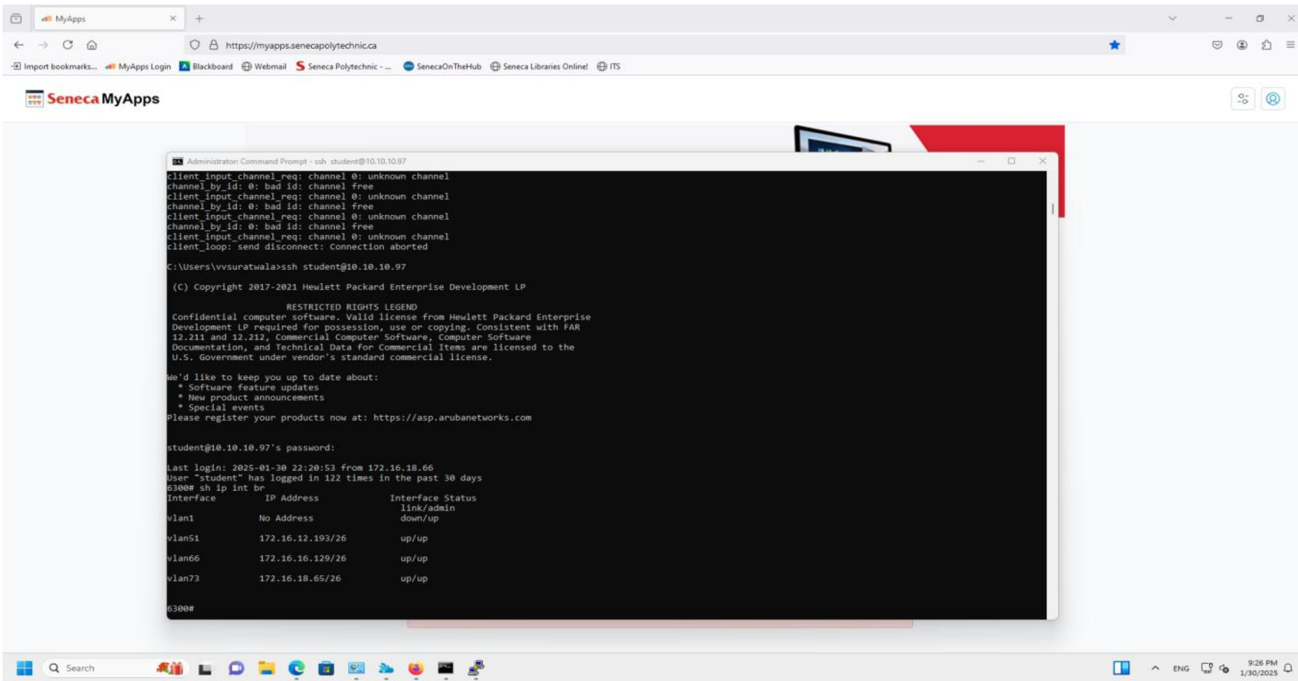


Client2 VM to Client3 VM

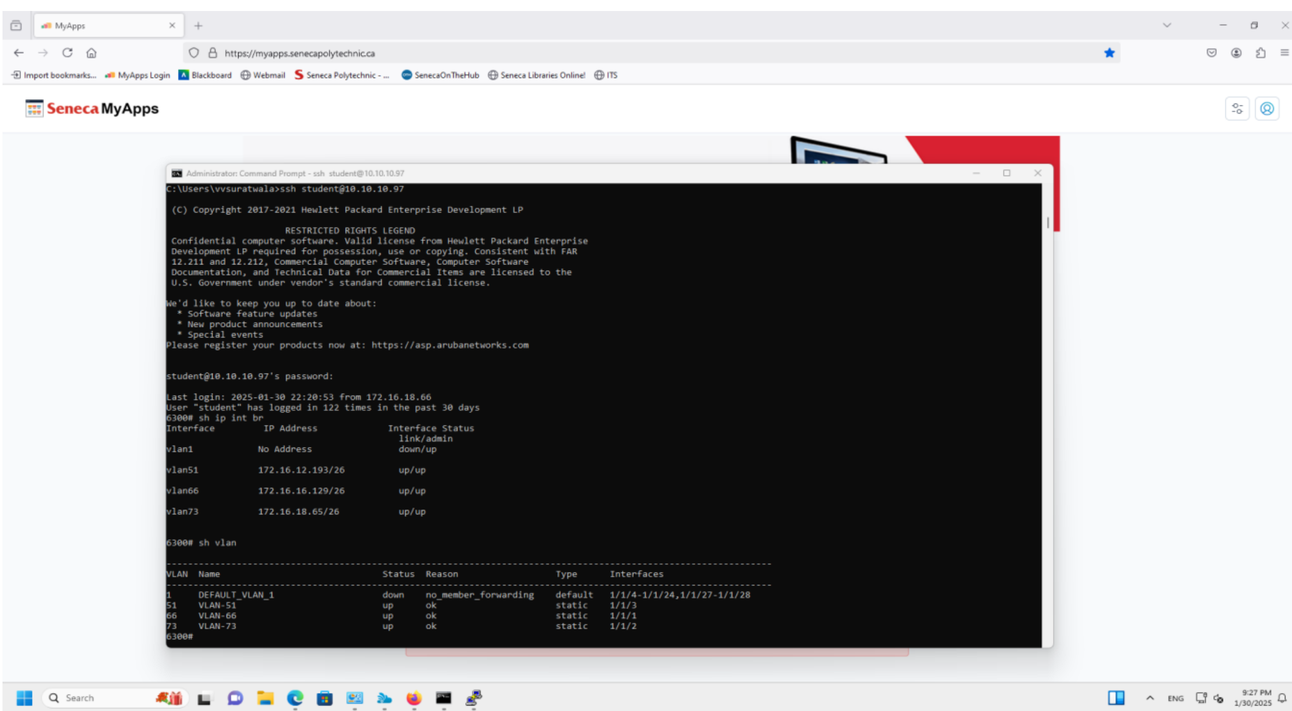


Appendix C: Commands on Switch

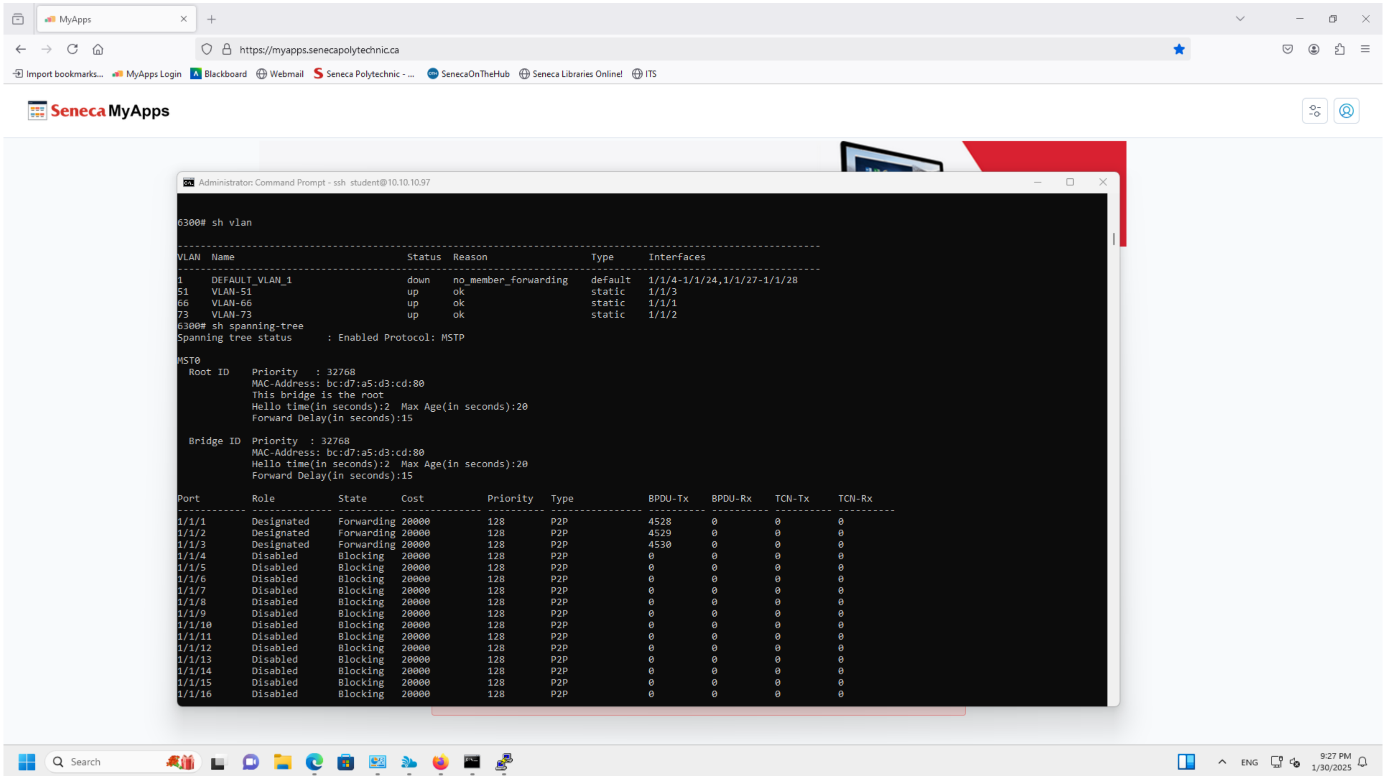
Sh ip int br



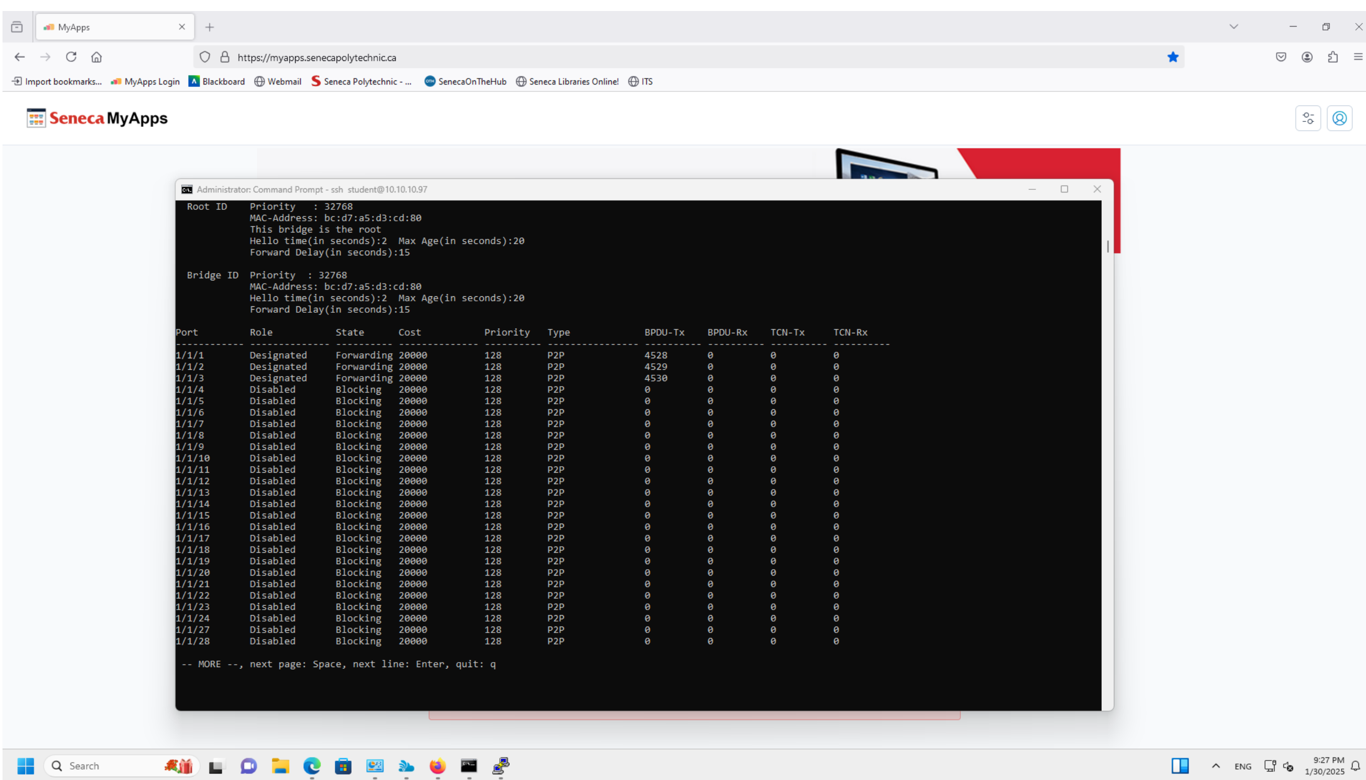
Sh vlan



Sh spanning-tree

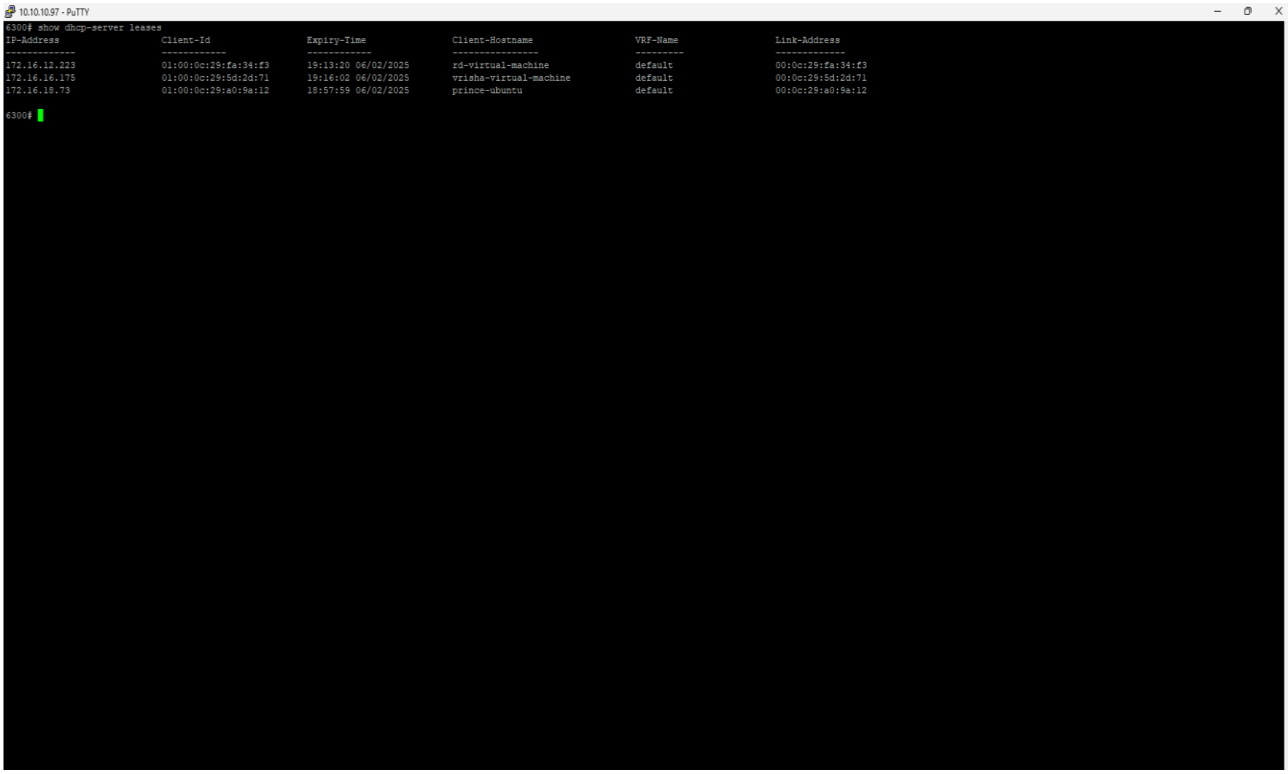


```
Administrator: Command Prompt - ssh  student@10.10.10.97

6300# sh vlan

--------------------------------------------------------------------------------------------------------
VLAN  Name                            Status  Reason                  Type      Interfaces
--------------------------------------------------------------------------------------------------------
1     DEFAULT_VLAN_1                  down    no_member_forwarding    default   1/1/4-1/1/24,1/1/27-1/1/28
51    VLAN-51                         up      ok                      static    1/1/3
66    VLAN-66                         up      ok                      static    1/1/1
73    VLAN-73                         up      ok                      static    1/1/2
6300# sh spanning-tree
Spanning tree status      : Enabled Protocol: MSTP

MST0
  Root ID    Priority   : 32768
             MAC-Address: bc:d7:a5:d3:cd:80
             This bridge is the root
             Hello time(in seconds):2  Max Age(in seconds):20
             Forward Delay(in seconds):15

  Bridge ID  Priority  : 32768
             MAC-Address: bc:d7:a5:d3:cd:80
             Hello time(in seconds):2  Max Age(in seconds):20
             Forward Delay(in seconds):15

Port         Role           State      Cost           Priority   Type             BPDU-Tx    BPDU-Rx    TCN-Tx     TCN-Rx
------------ -------------- ---------- -------------- ---------- ---------------- ---------- ---------- ---------- ----------
1/1/1        Designated     Forwarding 20000          128        P2P              4528       0          0          0
1/1/2        Designated     Forwarding 20000          128        P2P              4529       0          0          0
1/1/3        Designated     Forwarding 20000          128        P2P              4530       0          0          0
1/1/4        Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/5        Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/6        Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/7        Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/8        Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/9        Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/10       Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/11       Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/12       Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/13       Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/14       Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/15       Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/16       Disabled       Blocking   20000          128        P2P              0          0          0          0
```



```
Administrator: Command Prompt - ssh  student@10.10.10.97
  Root ID    Priority   : 32768
             MAC-Address: bc:d7:a5:d3:cd:80
             This bridge is the root
             Hello time(in seconds):2  Max Age(in seconds):20
             Forward Delay(in seconds):15

  Bridge ID  Priority  : 32768
             MAC-Address: bc:d7:a5:d3:cd:80
             Hello time(in seconds):2  Max Age(in seconds):20
             Forward Delay(in seconds):15

Port         Role           State      Cost           Priority   Type             BPDU-Tx    BPDU-Rx    TCN-Tx     TCN-Rx
------------ -------------- ---------- -------------- ---------- ---------------- ---------- ---------- ---------- ----------
1/1/1        Designated     Forwarding 20000          128        P2P              4528       0          0          0
1/1/2        Designated     Forwarding 20000          128        P2P              4529       0          0          0
1/1/3        Designated     Forwarding 20000          128        P2P              4530       0          0          0
1/1/4        Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/5        Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/6        Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/7        Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/8        Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/9        Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/10       Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/11       Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/12       Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/13       Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/14       Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/15       Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/16       Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/17       Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/18       Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/19       Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/20       Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/21       Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/22       Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/23       Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/24       Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/27       Disabled       Blocking   20000          128        P2P              0          0          0          0
1/1/28       Disabled       Blocking   20000          128        P2P              0          0          0          0

 -- MORE --, next page: Space, next line: Enter, quit: q
```

Sh dhcp-server leases



Appendix D: Switch Script Commands

```
conf ter

vlan 66,73,51
spanning-tree

int 1/1/1
no shutdown
no routing
vlan access 51
exit

int 1/1/2
no shutdown
no routing
vlan access 73
exit

int 1/1/3
no shutdown
no routing
vlan access 66
exit

vlan 66
name VLAN-66
exit

vlan 73
name VLAN-73
exit

vlan 51
name VLAN-51
exit

interface vlan 1
ip dhcp

interface vlan 51
ip address 172.16.12.193/26
no shutdown
```

```
exit

interface vlan 73
ip address 172.16.18.66/26
no shutdown
exit

interface vlan 66
ip address 172.16.16.130/26
no shutdown
exit

dhcp-server vrf default
pool VLAN-66
range 172.16.16.130 172.16.16.190 prefix-len 26
default-router 172.16.16.129
exit
pool VLAN-51
range 172.16.12.194 17.16.12.254 prefix-len 26
default-router 172.16.12.193
exit
pool VLAN-73
range 172.16.18.66 172.16.18.126
default-router 172.16.18.65
exit
enable
```